Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice

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We present structurally sparse qubit Hamiltonians for the nuclear shell model, constructed by applying the GeoVac angular-sparsity framework to the harmonic oscillator (HO) single-particle basis with a Mayer–Jensen spin–orbit term. The HO shell closures $2, 8, 20, 40, 70, 112$ follow from state counting on the oscillator labels, and the physical magic numbers $2, 8, 20, 28, 50, 82, 126$ are recovered with a spin–orbit strength $v_{ls}/\hbar\omega \approx 0.17$ and a Nilsson $\ell(\ell+1)$ correction $d_{\ell\ell}/\hbar\omega \approx 0.02$. We build a two-species qubit encoding for the deuteron ($1p + 1n$, 16 qubits, 688 non-identity Pauli terms, 1-norm ≈ 384 MeV) using the Minnesota nucleon–nucleon potential and Moshinsky–Talmi brackets, and extend the architecture to helium-4 ($2p + 2n$, 16 qubits, 904 Pauli terms) with intra-species antisymmetrization and proton–proton Coulomb repulsion; each encoded Hamiltonian reproduces its full-CI spectrum in the physical particle-number sector to 2×10^{-12} MeV. He-4 has $1.31\times$ the deuteron's Pauli terms, the added proton–proton and neutron–neutron channels; its $12.25\times$ larger Hilbert space costs no terms, because at fixed qubit count the Jordan–Wigner term set does not depend on the particle number. A composed nuclear–electronic encoding (26 qubits), a deuteron nuclear block coupled to a hydrogen-like electron through a proton-spin hyperfine term, exercises the multi-block architecture and quantifies the $\sim 10^{13}$ coefficient ratio between nuclear, electronic, and hyperfine scales. We state and apply the Fock projection rigidity theorem: the S^3 conformal projection that makes the one-electron Coulomb Hamiltonian equivalent to the S^3 Laplace–Beltrami operator (whose harmonics the GeoVac lattice's nodes label) requires the hydrogenic spectrum, and does *not* extend to HO or Woods–Saxon systems. A finite nucleus breaks it in a controlled way: the labels survive, n is released from being a whole number, and the s-wave quantum defect $-(2/5)(ZR)^2$ is the same for every n and continues across threshold, moments of the nuclear charge being consumed as the matching coefficients of an effective theory; a finite magnetization is priced the same way, the Zemach moment first, with a logarithm set by the charge alone; in the Dirac equation near and past $Z\alpha = 1$ the radius becomes compulsory (Sec. III A). For nuclear Hamiltonians, the GeoVac contribution is therefore the angular machinery and qubit encoding, not the conformal projection. All nuclear physics inputs (Minnesota strengths and ranges, used without the potential's space-exchange mixture; spin–orbit strength, $\hbar\omega$) follow the standard shell-model literature; the quantitative binding energies at $N_{\text{shells}} = 2$ are far from experiment and should be read as encoding-validation benchmarks, not as a nuclear-structure calculation.

I. INTRODUCTION

Nuclear quantum simulation is an active frontier in quantum computing [8–12]. Exact solvers (no-core shell model, NCSM [13, 14]) remain the gold standard for light nuclei $A \lesssim 16$, but scale factorially with the number of active nucleons, and their extension to medium-mass nuclei requires importance truncation or symmetry-adapted bases. On current and near-term quantum hardware, the practical bottleneck is the resource cost of the qubit Hamiltonian: the nuclear interaction is short-ranged and contains tensor and three-body components that produce dense Pauli representations in conventional encodings. Resource estimates for ab initio nuclear simulation using second-quantized Hamiltonians therefore sit at the edge of current hardware feasibility.

The GeoVac framework [34, 37] exploits angular-momentum selection rules to produce structurally sparse qubit Hamiltonians for molecular electronic structure, with Pauli counts that, at equal qubit count, range from above minimal-basis STO-3G to two orders of magnitude below larger Gaussian bases (Paper 20 [36]; a matched-qubit, not matched-accuracy, comparison). The angular selection rules, which are derived from Wigner $3j$ symbols and the Gaunt integral, are *independent of the radial potential*: they depend only on the single-particle angular basis, not on whether the interaction is Coulombic, Yukawa, or nucleon–nucleon. This paper investigates whether that sparsity carries over to nuclear shell model Hamiltonians.

We report three results. First, the harmonic oscillator shell closures and the Mayer–Jensen magic numbers are reproduced on the GeoVac $(n_r, \ell, m_\ell, \sigma)$ graph via an explicit spin–orbit term. Second, the deuteron and He-4 qubit Hamiltonians, built from the Minnesota NN potential using Moshinsky–Talmi brackets, reproduce their full-CI spectra — the deuteron has 16 qubits and 688 Pauli terms, and He-4 the same 16 qubits and 904, the difference being the proton–proton and neutron–neutron channels (the Hilbert-space dimension, $12.25\times$ larger for He-4, does not enter the term count at fixed qubit number). Third, a composed nuclear–electronic encoding, a deuteron nuclear block with a hydrogen-like electron, demonstrates that the block architecture used for core–valence molecular systems extends to nucleon–electron systems, though the 10^{13} coefficient ratio between nuclear-scale and hyperfine terms makes a single-pass quantum simulation impractical without block par-

titioning.

We also state the Fock projection rigidity theorem, which identifies the parts of the GeoVac framework that transfer to non-Coulomb systems (the angular machinery) and the parts that do not (the S^3 conformal projection). This delineation is the main conceptual contribution of the paper.

The paper is organized as follows. Section II constructs the nuclear shell model on the GeoVac lattice. Section III states the Fock projection rigidity theorem. Sections IV and V present the deuteron and He-4 qubit Hamiltonians. Section VI presents the composed nuclear–electronic encoding. Sections VIII and IX conclude.

II. NUCLEAR SHELL MODEL ON THE HYPERSPHERICAL LATTICE

A. Harmonic oscillator basis and shell closures

The standard starting point for the nuclear shell model is the three-dimensional isotropic harmonic oscillator with single-particle energies

$$E_N^{\mathrm{HO}} = \hbar\omega\left(N + \tfrac{3}{2}\right), \qquad N = 2n_r + \ell, \tag{1}$$

and degeneracy $(N+1)(N+2)$ per major shell including spin. Summing these degeneracies yields the HO shell closures

$$2,\ 8,\ 20,\ 40,\ 70,\ 112,\ \ldots$$

These are the correct magic numbers of a pure 3D isotropic oscillator but disagree with the experimentally observed nuclear magic numbers above $N = 20$.

On the GeoVac lattice, each state $(n_r, \ell, m_\ell, \sigma)$ is a graph node with diagonal energy given by Eq. (1). Because the HO shell spacing is uniform in N, the HO magic numbers are recovered by simple state counting at any choice of kinetic coupling; they do not depend on the graph off-diagonal structure. This is a numerical confirmation of the packing counting rule from Paper 0 [31] as applied to the HO shell degeneracies.

B. Spin–orbit term and the real magic numbers

Mayer and Jensen [1, 2] recognized that the observed nuclear magic numbers $28, 50, 82, 126$ require an additional spin–orbit interaction whose sign is *opposite* to the atomic case: the nuclear spin–orbit coupling lowers the $j = \ell + 1/2$ sub-shell below $j = \ell - 1/2$. Adding the Nilsson $\ell(\ell+1)$ term, which mimics the flattening of the Woods–Saxon mean field relative to the pure HO, one writes

$$E(N, \ell, j) = \hbar\omega\left(N + \tfrac{3}{2}\right) - v_{ls}\,\langle \boldsymbol{\ell}\cdot\mathbf{s}\rangle - d_{\ell\ell}\,\ell(\ell+1), \tag{2}$$

with $\langle \boldsymbol{\ell}\cdot\mathbf{s}\rangle = \ell/2$ for $j = \ell + 1/2$ and $-(\ell+1)/2$ for $j = \ell - 1/2$.

TABLE I. Level ordering from Eq. (2) at $v_{ls}/\hbar\omega = 0.171$, $d_{\ell\ell}/\hbar\omega = 0.021$. Shell closures at the experimentally observed magic numbers are marked with $\star$.

Shell	Levels (in order)	Cumul.
1	$0s_{1/2}$	$2^\star$
2	$0p_{3/2},\ 0p_{1/2}$	$8^\star$
3	$0d_{5/2},\ 1s_{1/2},\ 0d_{3/2}$	$20^\star$
4	$0f_{7/2}$	$28^\star$
5	$1p_{3/2},\ 0f_{5/2},\ 1p_{1/2},\ 0g_{9/2}$	$50^\star$
6	$1d_{5/2},\ 0h_{11/2},\ 2s_{1/2},\ 0g_{7/2},\ 1d_{3/2}$	$82^\star$
7	$1f_{7/2},\ 0i_{13/2},\ 2p_{3/2},\ 0h_{9/2},\ 1f_{5/2},\ 2p_{1/2}$	$126^\star$

A two-parameter scan over $(v_{ls}, d_{\ell\ell})$ identifies the range of parameters for which all seven physical magic numbers $2, 8, 20, 28, 50, 82, 126$ are recovered simultaneously as gaps in the single-particle spectrum. With seven HO shells and the midpoint of the valid region we obtain

$$\frac{v_{ls}}{\hbar\omega} \approx 0.171, \qquad \frac{d_{\ell\ell}}{\hbar\omega} \approx 0.021, \tag{3}$$

which is consistent with the Mayer–Jensen parameters tabulated in standard shell-model references [3, 4]. These are nuclear-physics inputs; the GeoVac framework contributes nothing to their determination. The recovered gaps are not uniform in magnitude: within the physically meaningful region (cumulative occupancy $\leq$ 126; boundaries above this sit at the $n_{\max} = 7$ truncation edge, where the missing $N = 7$ intruder levels distort the spectrum), the six lower magic numbers $2, 8, 20, 28, 50, 82$ are the six *largest* gaps, while 126 appears as a clear but non-dominant gap ($\approx 0.11\,\hbar\omega$), tied with the known sub-shell boundary at 40 and below several intermediate sub-shell gaps $(14, 56, 104, 118)$ — the expected granularity of the two-parameter Mayer–Jensen form at this $n_{\max}$ (gap structure pinned by `tests/test_paper23_magic_gaps.py`). Table I gives the level ordering, with shell closures marked.

The spin–orbit operator is implemented directly in the (m_ℓ, m_s) basis via

$$\boldsymbol{\ell}\cdot\mathbf{s} = \ell_z s_z + \tfrac{1}{2}(\ell_+ s_- + \ell_- s_+),$$

producing an off-diagonal coupling between $(m_\ell, \uparrow)$ and $(m_\ell + 1, \downarrow)$ within each (n_r, ℓ) subspace. The two-body electron-repulsion-like integrals are unchanged because spin–orbit is a one-body operator. The net effect on single-particle Hamiltonian sparsity is a modest increase from pure diagonal to a block-tridiagonal structure within each (n_r, ℓ) submatrix; the overall qubit encoding cost is dominated by the two-body terms in all cases we consider.

III. THE FOCK PROJECTION RIGIDITY THEOREM

The GeoVac framework for electronic structure uses a conformal projection, originally due to Fock [15], that maps the bound-state spectrum of the hydrogen atom onto the free Laplacian on the three-sphere S^3. The projection is

$$p_0 = \sqrt{-2E_n} = Z/n,$$

and it works because the hydrogen spectrum is ℓ-degenerate within each principal quantum number n: all orbitals (n, ℓ, m) with the same n have the same energy. The conformal projection then identifies the radial Schrödinger equation on $\mathbb{R}^3$ with the Laplace equation on S^3, whose harmonics are labelled exactly by the (n, ℓ, m) lattice and on which the S^3 Laplacian acts as the diagonal SO(4) Casimir $n^2 - 1$ (Paper 7 [33]). The lattice's edge Laplacian is not that operator and does not converge to it: it preserves ℓ and splits into disconnected ℓ-blocks (Paper 7, corrected 2026-09-25 from "converges to the continuous S^3 Laplacian in the asymptotic limit").

For nuclear shell model Hamiltonians the projection fails, for reasons that are elementary but worth stating precisely. We summarize the result as

Theorem (Fock projection rigidity). *For the S^3 Fock projection $p_0 = \sqrt{-2E}$ to carry the bound spectrum of a one-particle central-field Hamiltonian $p^2/2 + V(r)$ onto the S^3 Laplace–Beltrami operator (the SO(4) Casimir n^2-1 on the (n, ℓ, m) labels), two properties are needed: (i) $E_{n\ell}$ independent of ℓ within each n, and (ii) $E_n \propto -1/n^2$. Up to an additive constant in V and the value of Z, the Coulomb potential $-Z/r$ is the only central potential with both, as a consequence of its hidden SO(4) symmetry (the Runge–Lenz vector). A deformation $V(r) = -Z/r + \delta V(r)$ that is not a constant or a change of Z violates (i) or (ii), and the conformal equivalence between the radial Schrödinger problem and the S^3 Laplace–Beltrami operator is lost.*

[PRIOR ART] The result is classical: Fock's stereographic construction in momentum space [15] uses the hydrogenic relation $E_n \propto -1/n^2$ with no ℓ-dependence, and its group theory is reviewed by Bander and Itzykson [16]; it is stated here for its consequence, not proved anew. The consequence for the GeoVac lattice is that the diagonal node weights, which depend only on n in the Coulomb case (as the exact S^3 operator on the labels, the SO(4) Casimir $n^2 - 1$, does), would have to carry an ℓ-dependent correction; the labels then index a spectrum with no S^3 counterpart.

The relevant central potentials for nuclear physics — the 3D harmonic oscillator, the Woods–Saxon well, the square well, and the Yukawa potential — all have ℓ-dependent spectra. The 3D harmonic oscillator has its own accidental degeneracy (the SU(3) group of the isotropic oscillator groups states with common $N = 2n_r + \ell$), but the degeneracy pattern is different from Coulomb's SO(4), and no stereographic projection onto a compact manifold produces the S^3 Laplace–Beltrami structure that underlies the GeoVac electronic framework. The Bargmann–Segal transform [17] is the natural analogue for the HO, mapping the L^2 Hilbert space to a holomorphic reproducing-kernel space on $\mathbb{C}^3$. Paper 24 [38] builds the corresponding discrete label lattice on S^5; as on S^3, the HO spectrum is carried on the labels (diagonal, by the number operator), and the lattice's node Laplacian $L_0 = D - A$ is spectrally inert, so no graph-Laplacian convergence is involved or expected.

Negative result, positive implication. The rigidity theorem tells us what *cannot* be carried over from Papers 7 and 11–15 to the nuclear setting: the conformal projection, the identification of the (n, ℓ, m) labels with S^3 harmonics on which the S^3 Laplacian is the quantum kinetic operator in a dual description (a statement about the labels and the Casimir, not about the graph's edge Laplacian, which does not converge to Δ_{S^3}), and the $-1/16$ kinetic constant. What *can* be carried over is the angular machinery: the Wigner $3j$ coupling coefficients, the Gaunt integral selection rules, the block structure of the two-body interaction tensor, and the exactly Q-linear composed Pauli count that follows [37]. For nuclear shell models, we use the angular machinery to assemble the qubit Hamiltonian and use explicit radial matrix elements (Slater-like integrals computed from HO wavefunctions) for the radial part of the two-body interaction. The radial part is no longer related to the graph; it is a precomputed numerical table.

A. A finite nucleus: n released from integrality

The rigidity theorem says the S^3 projection needs exactly $-Z/r$. A nucleus of finite size changes the potential inside it, so $SO(4)$ must break. This subsection records what breaks and what survives, for a uniformly charged sphere of radius R (atomic units, infinite nuclear mass). The nucleus enters as a second length scale, the situation of the composed register of Sec. VI at two focal lengths; here it is read on the electron's labels alone. Inside the sphere the potential is $-Z(3R^2 - r^2)/(2R^3)$, an isotropic oscillator, so the electron's radial solution there is a Kummer function of r^2; outside it is a Coulomb function. Every level and phase shift below is one matching of the two at $r = R$. That interior oscillator is the electron's, not the nucleons' of Sec. II; the shared form is noted, not read.

[MEASURED] The labels l and m and the Gaunt rules survive, since the nucleus is round, and the radial node count stays $n-l-1$ (checked at $ZR = 0.01$ for eight states). What is released is n being a whole number: $E_{nl} = -Z^2/(2(n - \delta_{nl})^2)$, with

$$\delta_{nl} = -c_l\,(ZR)^{2l+2}\prod_{s=1}^{l}\Big(1-\frac{s^2}{n^2}\Big)+\cdots, \qquad c_l = \frac{2^{2l+2}\,k_l}{\big((2l+1)!\big)^2}, \quad k_l = \frac{2}{2} \tag{4}$$

so $c_0 = 2/5$, $c_1 = 1/105$ and $c_2 = 1/28350$. **[SYMBOLIC PROOF + MEASURED]** The leading term is first-order perturbation theory with the hydrogenic normalization near the origin, whose product over s is the one behind the continuum factor of Paper 7 [33]; exact matching at 50–60 digits, for ZR from 10^{-3} to 3.2×10^{-2} and $n \le 40$, confirms it to 10^{-10} for $l = 0$ and 10^{-11} for $l = 1, 2$. The defect depends on Z and R only through ZR. At leading order the s-wave defect is the same for every n, $-(2/5)(ZR)^2 = -(2/3)Z^2\langle r^2\rangle$: the Foldy–Friar contact term [22] read as a quantum defect. Beyond leading order the threshold defect, $n \to \infty$, follows exactly from the zero-energy matching, since the defect at threshold is the zero-energy phase shift over π (Seaton's limit, below). Inside, with $s = r/R$, the radial equation is $u'' = -x(3-s^2)\,u$, a power series in $x = ZR$; outside, the zero-energy pair is $\sqrt{2r}\,J_1(\sqrt{8r})$ and $-\sqrt{2r}\,Y_1(\sqrt{8r})$. Matching gives **[SYMBOLIC PROOF + MEASURED]**

$$\delta_{n0} = -\tfrac{2}{5}x^2 + \tfrac{32}{63}x^3 - x^4\Big[\tfrac{4}{25}\ln(2x) + \tfrac{8}{25}\gamma_E + \tfrac{28477}{225225}\Big] + O(x^5\ln x), \qquad x = ZR, \tag{5}$$

the threshold defect, $n \to \infty$, agreeing with the exact zero-energy matching to 1.3×10^{-18} at $x = 10^{-3}$ (the series is kept through x^5). Through x^3 it is the same at every n. At x^4 it is not:

$$\delta_{n0} - \delta_{\infty0} = -\delta_0^2\Big[\psi(n) - \ln n + \frac{1}{2n}\Big] - \frac{Z^4\langle r^4\rangle}{15\,n^2} + O(x^5\ln x), \tag{6}$$

with $\delta_0 = -\frac{2}{5}x^2$ the leading term and $\langle r^4\rangle = 3R^4/7$ for the sphere. **[PRIOR ART]** This is the $(Z\alpha)^6$ finite-size energy of Pachucki, Patkóš and Yerokhin [28] (their Eq. (30)) read as a quantum defect. **[MEASURED]** Exact matching at 60 digits reproduces the x^4 coefficients of $\delta_{10} - \delta_{20}$ and $\delta_{10} - \delta_{30}$ to 5×10^{-9}, with no $x^4\ln x$ term (pre-registered, `docs/predictions/2026-10-19_eft_checks.md`). The x^3 term is the Friar term $\frac{1}{3}Z^3\langle r^3\rangle_{(2)}$, the third moment of the nuclear charge convolved with itself [22], which for the uniform sphere is exactly $32R^3/21$. The weight-one constants γ_E and $\ln 2$ enter at fourth order [28]. The coefficient of $\ln(ZR)$ there, $-\delta_0^2$, is fixed by the leading term; the constant beside it depends on the profile: $-\delta_0^2[\ln 2 + 2\gamma_E]$ plus a rational for the sphere, a single γ_E for an exponential charge [28].

Other profiles. The structure is not the sphere's. For a thin charged shell the same derivation gives exactly $-\frac{2}{3}x^2 + \frac{16}{15}x^3 - x^4[\frac{4}{9}\ln(2x) + \frac{8}{9}\gamma_E + \frac{127}{315}] + \cdots$: the leading term set by $\langle r^2\rangle = R^2$, the Friar term with $\langle r^3\rangle_{(2)} = 16R^3/5$, and the same fourth-order logarithm, coefficient $-\delta_0^2$, again beside $2\gamma_E$. For a Gaussian nucleus of width b, solved numerically, the leading term is $-2(Zb)^2$ to 10^{-9} and the third-order coefficient is the Friar value $32/(3\sqrt{\pi})\,(Zb)^3$ to 1.6×10^{-6}, so a $\pi^{-1/2}$ enters the defect from the profile; the fourth-order logarithm has the coefficient $-\delta_0^2$ to 0.2% **[MEASURED]**.

Across threshold the defect continues as one function of energy. Since $1 - s^2/n^2 = 1 + 2s^2E/Z^2$, the leading defect is a polynomial $\mu(E)$, and to first order the short-range phase shift relative to Coulomb is

$$\tan\Delta_l(E) = \frac{\pi\,\mu(E)}{1 - e^{-2\pi Z/k}}, \tag{7}$$

the factor being Paper 7's continuum factor. **[MEASURED]** At $ZR = 10^{-3}$ the phase shift divided by Eq. (7) equals the bound side's ratio to 10^{-6} at $k = 1/2$, 1 and 2; without the factor it misses by 4.4% at $k = 2$. At threshold the factor tends to one and Seaton's limit [19] holds: the bound defects extrapolated to $n \to \infty$ and the phase shift over π extrapolated to $k \to 0$ agree to 10^{-15}.

Near and past $Z\alpha = 1$. In the Dirac equation the point-nucleus $1s$ level is $W = c^2\gamma$ with $\gamma = \sqrt{1-(Z\alpha)^2}$, rest energy included. At $Z\alpha = 1$ it reaches zero, and past it $\gamma = i\sigma$ with $\sigma = \sqrt{(Z\alpha)^2 - 1}$: the point-Coulomb Dirac operator then needs one boundary number at the origin, and a finite nucleus supplies it. For the uniform sphere, with an exact interior power series matched to the exterior at $r = R$, **[MEASURED]**: (i) below $Z\alpha = 1$ the $1s$ shift scales as $R^{2\gamma}$, the local exponent agreeing with 2γ to 6×10^{-5} at $Z\alpha = 0.9$ and better at 0.3 and 0.6; (ii) at $Z\alpha = 1$, W tends to zero slower than any power of R, the local exponent falling from 0.22 to 0.067 over $R = 10^{-3}$ to 10^{-9}; (iii) with $R = 1.2\,(2.5Z)^{1/3}$ fm the $1s$ level reaches $W = -c^2$ at $Z_{\rm cr} = 169.71$; (iv) at $Z\alpha = 1.2$ the radii at which successive $s_{1/2}$ levels reach $W = -c^2$ form a geometric sequence, the ratios 114.377, 113.995, 113.992, 113.992 tending to $e^{\pi/\sigma} = 113.992$. So past $Z\alpha = 1$ the radius fixes the levels only through $\ln R$ modulo π/σ: the continuous scale invariance of the point problem is broken to a discrete one by one consumed number. The radius is compulsory here, not a correction. The π in the period is the phase period of $r^{\pm i\sigma}$, and σ is a function of the consumed α. **[PRIOR ART]** The critical charge near 170 and its dependence on the nuclear radius are classical [20, 21]. One electron, no radiative corrections. Pre-registered predictions and their scoring, including one missed threshold for (ii): `docs/predictions/2026-10-18_relativistic_nucleus.md`.

A finite magnetization. For a point nuclear moment the hyperfine factor of an s level is the Fermi contact density $|\psi_n(0)|^2$. A real nucleus also spreads its magnetization, a third length scale. For a spherical magnetization density ρ_M the dipolar part averages to zero over an s electron, so the factor becomes $H_n = \int \rho_M\,|\psi_n|^2\,d^3r\,/\,|\psi_n^{\rm pt}(0)|^2$, with fractional change $\delta_n^{\rm hf} = H_n - 1$. We take a uniformly charged sphere of radius R with a point, uniform-ball or thin-shell magnetization of the same radius, and write $x = ZR$. (i) **[SYMBOLIC PROOF + MEASURED]** At leading order $\delta_n^{\rm hf} = -2Zr_Z$, where r_Z is the Zemach radius, the mean distance between a charge point and a magnetization point [23]: $r_Z = 3R/4$, $36R/35$ and $6R/5$, so the coefficients of x are $-3/2$, $-72/35$ and $-12/5$. These hold to 10^{-10} for $n = 1, 2, 3$ (40-digit matching, twelve points in $x = 10^{-5}$ to 3×10^{-3}), the same for every n.

The two distributions compose at this order through one joint moment, which neither radius fixes alone. (ii) **[SYMBOLIC PROOF]** With a point charge, $\delta_n^{\rm hf}$ is an entire function of x, with no logarithm at any order; through x^6 each coefficient is a polynomial in $1/n^2$ ($n = 1$ to 6), the x^2 coefficient being $1 + 1/(5n^2)$ for the uniform-ball magnetization. (iii) **[MEASURED]** With the uniform charge, $\delta_n^{\rm hf}$ contains $\frac{4}{5}\,x^2 \ln x$ for all three magnetization profiles and $n = 1, 2, 3$, to 4×10^{-6}. The coefficient is -2 times the leading quantum defect of Eq. (5), so the logarithm comes from the charge distribution alone. **[PRIOR ART]** The coefficient, $\frac{4}{3}Z^2\langle r^2\rangle$, is Karshenboim's [25] (his Sec. 2.5); its statement as -2 times the leading defect is the one addition here. (iv) **[MEASURED]** The n-dependence enters at x^2 as $\frac{4}{5}[\psi(n) - \ln n - 1/n]$ plus a profile constant, read off $n = 1, 2$ and confirmed at $n = 3$; that form is Karshenboim's, published for every n [25, 26]. An $x^3 \ln x$ term is also present, its coefficient the same for every n: $\frac{4}{5}$ times the coefficient of x, plus a constant, $-64/63$, the same for all three profiles and $n = 1, 2, 3$ to 10^{-12} in a fit that includes the $x^4 \ln^2 x$, $x^5 \ln^2 x$ and $x^6 \ln^2 x$ terms the data carry, with no $x^3 \ln^2 x$ term (`docs/predictions/2026-10-22_x3_constant_c.md`). It is the value that iterating the contact terms suggests; the fit tests that value but does not derive it. **[PRIOR ART]** The leading term is Zemach's [23]; a distributed nuclear magnetization goes back to Bohr and Weisskopf [24]; the nuclear corrections in light hydrogenic atoms are computed by Friar and Payne [29]; Paper 34 [30] carries the Zemach term as a projection. Higher-order size corrections have been expressed through moments of the two distributions (*Eur. Phys. J. D*, 2003, doi:10.1140/epjd/e2003-00018-8). Non-relativistic, spin magnetization only, no recoil or radiative corrections. Pre-registered predictions and their scoring, including the unregistered $x^3 \ln x$ term: `docs/predictions/2026-10-19_zemach_door.md`.

The price. The release consumes numbers of the nucleus, and no label fixes them: the mean square radius at leading order, the Friar moment at the next, then $\langle r^4\rangle$ and a log-moment constant together at fourth order; for the hyperfine factor, the Zemach moment of the charge and magnetization together, then the magnetic radius and a joint constant. They carry whatever number class the profile carries (rational for the sphere and the shell, $\pi^{-1/2}$ for a Gaussian); no π enters from the electron's side at leading order. From fourth order a logarithm of ZR enters, its coefficient fixed by lower orders and a profile-dependent constant beside it ($\ln 2 + 2\gamma_E$ plus a rational for the sphere), the weight-one class of the two-centre seed set of Paper 18 [40]. So the labels survive and the moments are consumed: a released axis whose price is calibration data, as the two-focal-length picture of Sec. VII expects. **[PRIOR ART]** The mechanism is the matching statement of nonrelativistic QED [27]: a structured nucleus enters only through the coefficients of contact operators, one new number for each independent operator at each order, and each logarithm comes from iterating lower-order contacts, so lower orders fix its coefficient. Karshenboim computed the hyperfine terms this way [25, 26], and the energy terms through $(Z\alpha)^6$ are in [28]. The physics is textbook (Foldy; Friar [22]; Seaton [19]); the contribution is the accounting on the electron's labels. Guards: `tests/test_paper23_finite_nucleus_door.py`, `tests/test_paper23_nuclear_profiles.py`, `tests/test_paper23_relativistic_nucleus.py`, `tests/test_paper23_hyperfine_door.py`, `tests/test_paper23_defect_n4.py`.

IV. THE DEUTERON: A TWO-SPECIES NUCLEAR QUBIT HAMILTONIAN

A. System and interaction

The deuteron is a bound state of one proton and one neutron in the ${}^3S_1$ channel, with experimental binding energy 2.2246 MeV and an rms charge radius of ~ 2.1 fm. The large radius is a consequence of the small binding; the deuteron wavefunction extends well beyond the range of the nuclear force, and accurate binding-energy calculations require oscillator bases with many major shells. We use the Minnesota potential [5], a phenomenological soft-core central interaction composed of three Gaussian terms. We use it in the form

$$V_{\rm NN}(r) = V_R\, e^{-\kappa_R r^2} + V_T\, e^{-\kappa_T r^2}\, P_{S=1} + V_S\, e^{-\kappa_S r^2}\, P_{S=0}, \tag{8}$$

with $(V_R, V_T, V_S) = (200, -178, -91.85)$ MeV and $(\kappa_R, \kappa_T, \kappa_S) = (1.487, 0.639, 0.465)$ fm^{-2} from Ref. [5], and $P_{S=1}$, $P_{S=0}$ the spin-triplet and spin-singlet projectors. The published potential multiplies each term by the space-exchange mixture $\frac{1}{2}[u + (2-u)P^r]$; the form used here has no space-exchange factor, which is the $u = 2$ member of that family (full strength in odd as well as even relative partial waves), not the usual $u \approx 1$. The two-body matrix elements in the HO basis are computed by transforming the antisymmetrized two-nucleon wavefunction to relative and center-of-mass coordinates using Moshinsky–Talmi brackets [6, 7].

B. Two-species qubit encoding

Protons and neutrons are distinct fermion species for the purposes of antisymmetrization: Pauli exclusion applies within each species but not between them. We therefore allocate two separate Jordan–Wigner registers, one of size Q_p for the proton orbitals and one of size Q_n for the neutron orbitals, and combine them by tensor product. At $N_{\rm shells} = 2$ (major shells $N = 0$ and $N = 1$) each species has

$$Q_p = Q_n = \underbrace{2}_{0s_{1/2}} + \underbrace{6}_{0p} = 8,$$

TABLE II. Deuteron qubit Hamiltonian at $N_{\text{shells}} = 2$, $\hbar\omega = 10$ MeV. "Non-I" denotes the non-identity Pauli terms. Hilbert-space dimension is $\binom{Q_p}{1}\binom{Q_n}{1} = 64$.

Quantity	Value
N_{shells}	2
Qubits $Q = Q_p + Q_n$	$8 + 8 = 16$
Non-I Pauli terms	688
Z-only terms	80
XY-containing terms	608
1-norm (non-I)	≈ 383.9 MeV
Hilbert dim. $(1p + 1n)$	64

giving a total $Q = 16$ qubits. The single-particle states are labeled in the uncoupled (n_r, ℓ, m_ℓ, m_s) basis; the coupling to j-scheme is performed implicitly via the Minnesota matrix elements.

The one-body Hamiltonian is

$$h_1 = \hbar\omega(2n_r + \ell + \tfrac{3}{2})\,\delta_{ij},$$

diagonal in the HO basis. The two-body Hamiltonian is

$$\hat{V}_{\text{pn}} = \sum_{ijkl} V_{ijkl}\, a^\dagger_{p,i} a^\dagger_{n,j} a_{n,l} a_{p,k},$$

where V_{ijkl} is computed from Eq. (8) via Moshinsky–Talmi brackets. Because protons and neutrons are on different registers, each term factorizes as

$$a^\dagger_{p,i} a_{p,k} \otimes a^\dagger_{n,j} a_{n,l},$$

yielding a tensor-product Pauli string that has support on both registers.

C. Resource counts

Table II summarizes the qubit Hamiltonian at $N_{\text{shells}} = 2$, $\hbar\omega = 10$ MeV. The non-identity Pauli term count is 688, with a 1-norm of ≈ 384 MeV dominated by the HO diagonal and the Minnesota spin-triplet contribution. Of the 688 terms, 80 are Z-only (number-operator products) and the remaining 608 carry at least one X or Y, reflecting the off-diagonal angular couplings generated by the Minnesota operator.

D. Binding energy at $N_{\text{shells}} = 2$

Diagonalizing the $(Q_p = 8) \times (Q_n = 8) = 64$-dimensional FCI matrix at $\hbar\omega = 10$ MeV and subtracting the HO zero-point energy gives a bound-state energy that is far from the experimental -2.2246 MeV. This is expected: the deuteron's rms radius of ~ 2.1 fm is comparable to the HO length at $\hbar\omega = 10$ MeV, so $N_{\text{shells}} = 2$ has insufficient radial flexibility to describe the long tail of the wavefunction. Converged Minnesota calculations require $N_{\text{shells}} \gtrsim 6$–$8$. We therefore emphasize that the results in Table II are *encoding-validation* benchmarks, not nuclear-structure predictions. The Pauli count and 1-norm characterize the cost of the qubit Hamiltonian. The encoding is validated against the FCI matrix: the Pauli sum restricted to the 64 one-proton, one-neutron basis states has the FCI spectrum to 2×10^{-13} MeV, both by direct evaluation of the Pauli strings and through OpenFermion's operator construction (`tests/test_paper23_resource_counts.py`, `tests/test_nuclear_deuteron.py`). The off-diagonal one-body pieces $a^\dagger_{p,i} a_{p,k}$ that each two-body term factorizes into carry the Jordan–Wigner coefficients $\pm i/4$ on their XY and YX strings; these are kept complex through every product, and only the final Hermitian sum is real. Recent deuteron-targeted quantum-resource work [11, 12] provides direct comparison points at comparable qubit count for the encoding cost reported here.

V. HELIUM-4: ANTISYMMETRIZATION WITHIN SPECIES

We extend the deuteron encoding to He-4, a doubly magic nucleus with two protons and two neutrons, experimental binding energy 28.296 MeV. The architecture is the same as for the deuteron, with two changes:

1. Intra-species antisymmetrization: protons fill a 2-particle Slater determinant on the proton register, neutrons fill a 2-particle Slater determinant on the neutron register. The resulting two-body terms include pp and nn antisymmetrized matrix elements in addition to the pn cross-species terms.

2. Coulomb repulsion between the two protons, added as a conventional two-body Coulomb matrix element in the HO basis (approximately 0.6 MeV for the ground state).

Both changes leave the number of qubits unchanged ($Q = 16$) because adding particles does not add spin orbitals; they only increase the Hilbert-space dimension. At $N_{\text{shells}} = 2$ the pp pair occupies a 2-particle subspace of dimension $\binom{8}{2} = 28$, and similarly for nn, so the full He-4 Hilbert space has dimension $28 \times 28 = 784$.

Table III compares He-4 to the deuteron. The Pauli count grows by 31% (688 $\rightarrow$ 904), while the Hilbert-space dimension grows by a factor of $784/64 = 12.25$. The two numbers are not related: at fixed Q the Jordan–Wigner operator, and so its term set, does not depend on how many particles occupy the register, and the $12.25\times$ is the growth of the particle-number sector the operator is restricted to. The added terms are the pp and nn two-body channels, absent from the deuteron, which has one nucleon of each species. The Coulomb correction does not add any new Pauli terms (it modifies only the existing pp coefficients). As for the deuteron,

TABLE III. Comparison of deuteron and He-4 qubit Hamiltonians at $N_{\text{shells}} = 2$. Both systems use 16 qubits; the difference is the Hilbert-space dimension and the number of particles. Pauli counts refer to non-identity terms.

System	Q	Pauli	Hilbert dim.
Deuteron ($1p + 1n$)	16	688	64
He-4 ($2p + 2n$, no Coulomb)	16	904	784
He-4 ($2p + 2n$, + Coulomb)	16	904	784

the encoded He-4 Hamiltonian restricted to the 784 two-proton, two-neutron basis states has the FCI spectrum, to 2×10^{-12} MeV with and without Coulomb (`tests/test_paper23_resource_counts.py`).

The 1-norm of the He-4 Hamiltonian at $\hbar\omega = 10$ MeV is ≈ 641 MeV (without Coulomb) and ≈ 632 MeV (with Coulomb), about 70% of it in the diagonal (Z-only) terms. The ground-state energy from FCI diagonalization is far from the experimental -28.296 MeV at $N_{\text{shells}} = 2$, for the same reason as in the deuteron: the HO basis is too small to describe the α-particle charge distribution. Converged Minnesota calculations of He-4 require $N_{\text{shells}} \gtrsim 6$. As with the deuteron, we emphasize that Table III is an encoding-validation benchmark.

VI. COMPOSED NUCLEAR–ELECTRONIC DEUTERIUM

The composed architecture used for core–valence molecular electronic structure [34, 35] extends naturally to the nuclear–electronic case: the nuclear block is the deuteron Hamiltonian from Sec. IV, the electronic block is a single-electron hydrogen Hamiltonian on the (n, ℓ, m) lattice at $n_{\max} = 2$, and the coupling between them is given by two one-body-like operators. The couplings take proton inputs: the proton charge radius, and a hyperfine term coupling the spin of the proton's $0s$ orbital to the electron with the hydrogen constant. They are hydrogen's couplings placed on a deuteron nuclear block, not those of deuterium (nuclear spin $I = 1$, charge radius ≈ 2.13 fm, ground-state splitting 327.4 MHz), which the code does not build.

A. Blocks and couplings

Nuclear block (Block A). One proton and one neutron in the HO basis at $N_{\text{shells}} = 2$, 16 qubits, as described in Sec. IV. Energy scale $\sim 10^1$–10^2 MeV after conversion to atomic units.

Electronic block (Block B). A single electron in the hydrogenic $1s$–$2p$ manifold (Coulomb potential $-1/r$), $Q_e = 10$ qubits (5 spatial orbitals × 2 spins: $1s$, $2s$, $2p_{-1}$, $2p_0$, $2p_{+1}$). The ground state is the exact hydrogenic $1s$ at -0.5 Ha.

Finite-size coupling. A one-body operator on the electronic register shifts the $1s$ energy by the finite nuclear-size correction $\Delta E_{ns} = (2/3)Z^4\langle r^2\rangle/n^3$ [18, 22], which for a uniformly charged sphere of radius R is $(2/5)Z^4R^2/n^3$. At the proton rms charge radius $\langle r^2\rangle^{1/2} = 0.8414$ fm, the $1s$ shift is $\Delta E \approx 1.69 \times 10^{-10}$ Ha, far below the subtraction precision of float64 arithmetic when combined with the nuclear-scale terms. The shift is reported analytically rather than by numerical subtraction.

Hyperfine coupling. The spin $\mathbf{I}$ of the proton's $0s$ orbital couples to the electron spin $\mathbf{S}$ via $H_{\text{hf}} = A_{\text{hf}}\,\mathbf{I}\cdot\mathbf{S}$, where the hyperfine constant for the hydrogen 21-cm transition is $A_{\text{hf}} = 1420.4$ MHz $\approx 2.16 \times 10^{-7}$ Ha. The singlet–triplet gap is A_{hf}, and the maximum shift of any single ground-state manifold member is $3A_{\text{hf}}/4 \approx 1.62 \times 10^{-7}$ Ha.

The composed Hamiltonian is

$$H = H_{\text{nuc}} \otimes I_e + I_{\text{nuc}} \otimes H_e + V_{\text{fs}} + H_{\text{hf}}, \tag{9}$$

with V_{fs} the finite-size shift on the electronic register and H_{hf} the cross-register hyperfine operator.

B. Resource counts and coefficient hierarchy

The composed Hamiltonian uses $Q = 16 + 10 = 26$ qubits. The Pauli count on the full register is 710 non-identity terms, of which 688 come from the nuclear block, 10 from the electronic block diagonal (one Z per spin-orbital plus the absorbed finite-size shift), and 12 from the cross-register hyperfine operator. The finite-size shift is absorbed into the diagonal electronic terms and does not add new Pauli strings. Restricted to the one-proton, one-neutron, one-electron sector, the Pauli sum has the spectrum of the same Hamiltonian assembled directly as a matrix, to float64 roundoff (10^{-8} Ha on entries of 5×10^5 Ha; `tests/test_nuclear_electronic.py`).

The coefficient hierarchy spans ~ 13 orders of magnitude (Table IV): the nuclear diagonal dominates at $\sim 5 \times 10^5$ Ha (after conversion from MeV to Ha via 1 Ha ≈ 27.2 eV), the electronic binding is ~ 0.5 Ha, the hyperfine coefficient is $\sim 10^{-8}$ Ha per Pauli string (the $A_{\text{hf}}/16$ from the $\mathbf{I}\cdot\mathbf{S}$ decomposition), and the finite-size shift is $\sim 10^{-10}$ Ha but does not appear in the Pauli list because it is below the float64 subtraction floor relative to the nuclear-scale diagonal.

C. Multi-scale precision problem

The $\sim 10^{13}$ coefficient ratio is the central obstacle to running Eq. (9) as a single-pass quantum simulation on near-term hardware. To resolve a hyperfine shift of 10^{-7} Ha against a diagonal that contains 10^5 Ha entries requires an effective energy precision of $\sim 10^{-12}$ in relative terms, which is well beyond the scope of VQE or QPE

TABLE IV. Coefficient hierarchy in the composed nuclear–electronic deuterium Hamiltonian (atomic units).

Contribution	Scale (Ha)
Max Pauli coeff (nuclear)	$\sim 5 \times 10^{5}$
Electronic binding scale	$\sim 10^{-1}$
Hyperfine coeff (per Pauli)	$\sim 10^{-8}$
Finite-size shift (analytic)	$\sim 10^{-10}$
Pauli coeff ratio max/min	$\sim 4 \times 10^{13}$

at current shot counts. The finite-size shift is not only below VQE precision but below the float64 subtraction floor of the nuclear diagonal, and must be computed analytically as a perturbative shift rather than by direct diagonalization.

The practical consequence is that the composed nuclear–electronic Hamiltonian should be solved in block-partitioned form: the nuclear block is solved first (using the deuteron Hamiltonian in isolation), the electronic block is solved second (at fixed proton charge distribution), and the cross-block couplings are treated perturbatively as classical corrections. This is the quantum-chemistry analogue of the Born–Oppenheimer approximation, and it is appropriate whenever the block energies span orders of magnitude. The composed Hamiltonian in Eq. (9) is most useful as an *architectural* object: it lets one write down the full Hamiltonian on a single qubit register, verify correctness by comparison with the block decomposition, and reason about the cost of adding cross-block couplings, even if the single-pass solution is not practical.

D. Positioning in the noncommutative-geometry literature

The composed nuclear–electronic Hamiltonian of this section is also an explicit Connes-style spectral-triple-level construction in the sense of the GeoVac spectral-triple framework (Paper 32 [39] §V, §VIII.D). The Hilbert space is the tensor product of two real-space registers for two physically distinct quantum particles (proton + electron) at categorically different focal lengths; the Dirac operator carries cross-register coupling ($\mathbf{I} \cdot \mathbf{S}$ hyperfine); and the construction is audited against the canonical Connes axioms by the surrounding Paper 32 framework. The published noncommutative-geometry / spectral-triple lineage (Connes–Marcolli–Chamseddine SM almost-commutative geometry; Connes–van Suijlekom spectral truncations and their convergence theorems) does not contain a directly comparable real-space multi-particle construction: the SM case couples a Riemannian factor with a *finite-dimensional internal* factor, and the truncation-convergence line covers single triples or one-infinite-plus-one-finite products. The structural observation, an explicit literature comparison, and the empirical validation against the 21 cm hyperfine gap are formalized in a standalone Zenodo memo at `papers/group4_quantum_computing/track_ni_spectral_triple_z` with cross-references to Paper 32 §V (sub-sector table) and §VIII.D (frontier-of-field framing).

VII. CROSS-REGISTER V_{eN} AND OPERATOR-LEVEL MAGNETIZATION-DENSITY INNER-FLUCTUATIONS

The composed nuclear–electronic Hamiltonian of Sec. VI couples two real-space registers but treats both the electron–nucleus spatial interaction and the proton magnetization as classical scalars: the finite-size correction V_{fs} is evaluated at a fixed scalar R_p and absorbed into the electronic diagonal, and no operator on the proton register represents the nuclear coordinate. This section reports the construction of two operator-level upgrades that promote each scalar to a bilinear matrix element on the joint $\mathcal{H}_e \otimes \mathcal{H}_p$ register:

1. a cross-register Coulomb operator $V_{eN}(\hat{\mathbf{r}}_e, \hat{\mathbf{R}}_p)$ at distinct Sturmian focal lengths (λ_e, λ_n), built on the Roothaan 1951 closed form for the bilinear matrix element of $1/|\mathbf{r} - \mathbf{R}|$ between hydrogenic states [43];

2. a magnetization-density inner-fluctuation ω_{magn}, parameterized by a calibrated radial profile $\rho_M(r; r_Z)$ on the proton register, that reproduces the Eides Zemach correction [44] at the operator level rather than as an external scalar substitution.

The two operators share the same joint-register layout, the same diagonal-density Jordan–Wigner Pauli encoding, and the same multi-λ Shibuya–Wulfman radial machinery; they differ only in radial kernel ($-Z/|\mathbf{r}_e - \mathbf{R}_p|$ for V_{eN}, the magnetization profile ρ_M for ω_{magn}). They realize, on a finite truncation, the structural picture from Phase B of the multi-focal sprint sequence: V_{eN} and ω_{magn} are categorically distinct *components* of a single composed inner fluctuation $\omega^{\mathrm{comp}} = \omega_{\mathrm{recoil}} + \omega_{\mathrm{magn}} + \cdots$, structurally analogous to how the Connes–Chamseddine inner fluctuation $\omega = \omega_{\mathrm{gauge}} + \omega_{\mathrm{Higgs}}$ splits in the Standard Model spectral triple [39, 45].

The empirical content of this section is honest about its scope. V_{eN} reproduces the leading-order Bethe–Salpeter recoil correction to the hydrogen 1s energy at 2.86%, and follow-up symbolic analysis (Sec. VII B) shows that this residual is a basis-truncation artifact of the $1s \times 1s$ Sturmian representation, not a missing physical sub-leading correction. ω_{magn} reproduces the closed-form leading-order Zemach shift $-2Z\alpha\, m_e r_Z$ (Eides § 11.1.1), with its profile ρ_M calibrated so that the first radial moment is $M_1[\rho_M] = r_Z$; the agreement checks the operator construction, not the value of r_Z. The structural interpretation is that the framework computes the operator action

given a fixed ρ_M, and the calibration (r_Z value, profile family) remains a Layer-2 input from QCD/scattering data, consistent with the structural-skeleton scope pattern documented in CLAUDE.md §3.5.

A. Cross-register V_{eN} on the joint Sturmian register

The cross-register two-body matrix element

$$M = \langle \chi^{\lambda_e}_{n_e\ell_e m_e}(\mathbf{r}_e)\, \chi^{\lambda_n}_{n_n\ell_n m_n}(\mathbf{R}_p) \,|\, \frac{1}{|\mathbf{r}_e - \mathbf{R}_p|} \,|\, \chi^{\lambda_e}_{n'_e\ell'_e m'_e}(\mathbf{r}_e)\, \chi^{\lambda_n}_{n'_n\ell'_n m'_n}(\mathbf{R}_p) \rangle \tag{10}$$

is a six-coordinate integral that closes in elementary functions when the proton coordinate $\mathbf{R}_p$ is promoted from a classical scalar to a quantum operator on a Sturmian register at focal length λ_n. The closure has three pieces:

1. Multipole expansion of $1/|\mathbf{r}-\mathbf{R}|$ in Legendre polynomials with bilateral Gaunt selection on both the electron and the proton angular labels. The expansion terminates exactly at $L_{\max} = \min(\ell_e+\ell'_e,\, \ell_n+\ell'_n)$ — the tighter of the two Gaunt bounds.
2. A bilateral angular factor: a product of two Wigner $3j$ symbols, one on each side, with cross-register total-m conservation $m_e + m_n = m'_e + m'_n$.
3. A double radial integral whose inner R-integral at fixed r is the multi-λ Shibuya–Wulfman split-region integral with combined decay rate $\alpha_{\rm tot} = \lambda_a + \lambda_b$, evaluated analytically through incomplete gamma functions, and whose outer r-integral is performed by Gauss–Laguerre quadrature or, for the simplest case, by a closed form.

For $(n_e,\ell_e) = (n'_e,\ell'_e) = (1,0)$ and $(n_n,\ell_n) = (n'_n,\ell'_n) = (1,0)$, $L = 0$, the integral collapses to the Roothaan 1951 closed form

$$\boxed{J_0(\lambda_e,\lambda_n) = \frac{\lambda_e\lambda_n\big(\lambda_e^2 + 3\lambda_e\lambda_n + \lambda_n^2\big)}{(\lambda_e+\lambda_n)^3}} \tag{11}$$

which reproduces the textbook value $5/8$ at $\lambda_e = \lambda_n = 1$, is symmetric in $(\lambda_e \leftrightarrow \lambda_n)$, and recovers the point-nucleus limit $J_0 \to \lambda_e$ as $\lambda_n \to \infty$ at the rate $O(1/\lambda_n^2)$.

a. Implementation. The construction is implemented in `geovac/cross_register_vne.py` ($\sim$ 1330 lines, 56 fast tests) as an additive extension of the single-λ Shibuya–Wulfman module of Paper 19. Two new low-level functions in `geovac/shibuya_wulfman.py` carry the multi-λ generalization: `_hydrogenic_poly_coeffs_lam(lam, n, l)` returns the Sturmian polynomial coefficients $R^{\lambda}_{n\ell}(r) = e^{-\lambda r}\sum_k c_k r^k$ at independent exponent λ, and `_radial_split_integral_lam(lam_a, n_1, l_1, lam_b, n_2, l_2, L, R)` computes the mismatched-exponent split-region radial integral. Bit-identical regression to the legacy single-λ functions is verified to relative error $\sim 10^{-12}$ across the full single-λ test suite (26/26 tests pass without modification).

b. Verification. The four structural properties expected of a cross-register Coulomb operator are verified in `tests/test_cross_register_vne.py`: hermiticity ($V[i,j,k,l] = V[k,l,i,j]^*$), block-diagonal structure under cross-register total-m conservation, monotonic convergence of $J_0(\lambda_e,\lambda_n)$ to λ_e as $\lambda_n \to \infty$ (verified at $\lambda_n \in \{10, 10^2, 10^3, 10^4, 10^5\}$ with relative error scaling as $2/\lambda_n^2$), and bit-identical reproduction of the bare-Coulomb V_{eN} in the limit $\lambda_n \to \infty$. Multipole termination is verified at $L_{\max} = \ell_e + \ell'_e$ across $\ell_e, \ell'_e \in \{0,1,2\}$, preserving the tight Gaunt bound under exponent mismatch.

B. Bethe–Salpeter leading order and the basis-truncation clarification

The cross-register V_{eN} at $\lambda_e = 1$, $\lambda_n = 2\sqrt{M_p/m_e} \approx 85.7\,\mathrm{bohr}^{-1}$, applied to the hydrogen 1s ground state, produces a recoil-like shift

$$\Delta E^{\rm cross}_{\rm recoil} = -Z\big[J_0(\lambda_e,\lambda_n) - \lambda_e\big] = +2.65\times10^{-4}\,\mathrm{Ha}, \tag{12}$$

which agrees with the textbook leading-order Bethe–Salpeter recoil $|E_1|\cdot m_e/m_p = +2.72\times10^{-4}$ Ha at the 2.86% level. The same numerical agreement appears against the leading-order term of the Pachucki–Patkóš–Yerokhin 2018 Foldy–Wouthuysen-reduced two-particle Hamiltonian [46].

The value $\lambda_n = 2\sqrt{M_p/m_e}$ is calibrated: it is fixed by requiring the leading $O(1/\lambda_n^2)$ Roothaan term in Eq. (11) to match the reduced-mass leading-order shift,

$$+\frac{2Z\lambda_e}{\lambda_n^2} \stackrel{!}{=} \frac{m_e}{m_p}\,|E_1| = \frac{1}{2M_p} \quad\Longrightarrow\quad \lambda_n = 2\sqrt{M_p/m_e}.$$

The proton's quantum-mechanical zero-point spread in the reduced-mass coordinates is encoded as the Sturmian focal length on the nucleus register, in direct analogy with the use of $\sqrt{Z/n}$ on the electron register.

The 2.86% residual deserves a sharper interpretation than "higher Roothaan terms." Symbolic Taylor expansion of Eq. (11) in $\varepsilon = 1/\lambda_n$ at $\lambda_e = 1$ yields

$$J_0(1,1/\varepsilon) = 1 - 2\varepsilon^2 + 5\varepsilon^3 - 9\varepsilon^4 + 14\varepsilon^5 - 20\varepsilon^6 + 27\varepsilon^7 - 35\varepsilon^8 + \cdots \tag{13}$$

with magnitudes $\{2,5,9,14,20,27,35,\ldots\}$ growing quadratically. At calibrated $\lambda_n = 2\sqrt{M_p/m_e}$, the order-$k$ contribution scales as $1/\lambda_n^k = (m_e/m_p)^{k/2}/2^k$, so *odd* k produces a half-integer power of m_e/m_p while *even* k produces an integer power.

Half-integer powers of m_e/m_p are structurally absent from the Pachucki–Patkóš–Yerokhin Foldy–Wouthuysen expansion of the two-particle Hamiltonian: their series is a polynomial in the single small parameter m_e/m_p with integer-power-only terms. The Roothaan odd-k tower is therefore not a sub-leading recoil correction at all, but a

basis-truncation artifact of the $1s\times1s$ Sturmian representation at fixed $n_{\max} = 1$. Numerically, the half-integer tower at $\lambda_n = 85.7$ sums to -7.95×10^{-6} Ha, exactly -2.92% of the Bethe–Salpeter leading order, accounting for nearly all of the 2.86% drift.

The complementary integer-only sub-sum, $\sum_{k=2,4,6,8,\ldots} c_k/\lambda_n^k$, lands at $+2.72475 \times 10^{-4}$ Ha, a $+0.061\%$ deviation from Bethe–Salpeter. We flag this sub-percent agreement as *fortuitous*, not structural: the order-$k = 4$ Roothaan coefficient is $+9$, while the physical Pachucki next-order coefficient is $-1/(2M_p^2)$ — the magnitudes are comparable but the *signs are opposite*. The integer-only filter accidentally lands close because $1/\lambda_n^4$ is small ($\sim 10^{-7}$ Ha) at the calibrated focal length, not because the coefficients are correct. At larger Z where the next-order recoil grows as Z^5, the fortuitous cancellation degrades.

The structural conclusion is sharper than the numerical one: the cross-register V_{eN} at $n_{\max} = 1$ reproduces the leading order of Pachucki 2018 by the calibration of λ_n, with sub-leading content *basis-truncation-bounded*; closing the next-order sign disagreement structurally requires multi-shell expansion to $n_{\max} \geq 2$ on both registers, which is named as the natural follow-up but is out of scope for this paper. The single-parameter calibration $\lambda_n = 2\sqrt{M_p/m_e}$ is structurally a leading-order-only fix; matching higher orders requires more basis functions, which is consistent with the Layer-2 calibration interpretation of focal length in Paper 18 §IV [40].

C. Operator-level magnetization-density inner-fluctuation $\omega_{\rm magn}$

The Eides Zemach correction to the hydrogen hyperfine splitting [44] is conventionally written as a scalar substitution

$$\frac{\Delta\nu_Z}{\nu_F} = -2Z\alpha m_e\, r_Z, \qquad r_Z = \int d^3r\, d^3r'\, \rho_E(r')\, \rho_M(|\mathbf{r}-\mathbf{r}'|), \tag{14}$$

where ρ_E and ρ_M are the unit-normalized proton charge and magnetization densities and r_Z is the first moment of their convolution. Sprint HF-4 (Sec. VI, footnote; Sprint HF [50] for the original 21 cm precision test) closed the framework's residual to the 21 cm transition by treating r_Z as a Layer-2 external input scalar; that closure landed at $+18$ ppm, consistent with the Eides Ch. 11 budget for multi-loop QED plus nuclear polarizability.

The construction reported here promotes the scalar substitution to an operator-level inner-fluctuation on the joint $\mathcal{H}_e \otimes \mathcal{H}_p$ register. The operator

$$\omega_{\rm magn} = -2Zm_e \int d^3R\, \rho_M(R)\, \hat{r}_Z[R] \tag{15}$$

is built from a unit-normalized radial profile $\rho_M(r; r_Z)$ that is itself parameterized by the Zemach radius as a Layer-2 calibration scalar. The bilinear matrix element on the joint register at the leading multipole $L = 0$ collapses to

$$\langle \hat{r}_Z \rangle = \int d^3R\, \rho_M(R)\, R + O(r_Z/a_0)^2 = M_1[\rho_M] + O(r_Z/a_0)^2 = r_Z + O \tag{16}$$

The Eides scalar substitution is recovered because the profile is calibrated to it: ρ_M is chosen with first radial moment equal to r_Z (standing in for the charge–magnetization convolution of Eq. (14)), and the operator-level construction returns that moment through the standard hydrogenic 1s contact integral. Higher-order corrections in r_Z/a_0 are the second moment $\langle r^2 \rangle_{\rho_M}$ and beyond, suppressed by $\sim 7.8 \times 10^{-6}$ relative to the leading shift on hydrogen.

a. Implementation. The module `geovac/magnetization_density.py` ($\sim$ 480 lines, 27 fast tests) supports three radial profiles: Gaussian $\rho_M(r) = (\beta/\pi)^{3/2}\, e^{-\beta r^2}$ with $\beta = 4/(\pi r_Z^2)$, exponential $\rho_M(r) = (\kappa^3/8\pi)\, e^{-\kappa r}$ with $\kappa = 3/r_Z$, and a delta limit. All three are unit-normalized ($M_0 = \int d^3r\, \rho_M = 1$) and calibrated to the same first moment ($M_1 = r_Z$ for Gaussian and exponential by construction). They differ only at the second moment M_2, which controls the sub-leading shape-dependent correction.

b. Verification. At the Eides calibration $r_Z = 1.045\,\text{fm} = 1.974\times10^{-5}$ bohr, the operator-level construction gives

$$\left.\frac{\Delta\nu_Z}{\nu_F}\right|_{\rm operator} = -39.495\,\text{ppm}, \tag{17}$$

which equals the closed-form leading Zemach term $-2Z\alpha\, m_e r_Z$ (Eides § 11.1.1) at $r_Z = 1.045\,\text{fm}$ to the printed digits. Profile independence at leading order is verified to machine precision: Gaussian and exponential profiles, both calibrated to the same r_Z, produce identical leading-order shifts. They differ at order 2 by $\sim 13\%$ (Gaussian $M_2 = 4.59 \times 10^{-10}$ bohr^2 vs. exponential $M_2 = 5.20 \times 10^{-10}$ bohr^2), well below any precision target relevant to hydrogen 21 cm.

c. Composition with V_{eN}. On a minimal joint register ($1s_e \times 1s_p$, $Q = 2$ qubits), both V_{eN} and $\omega_{\rm magn}$ produce four-string Pauli sums with shared support on $\{II, ZI, IZ, ZZ\}$. The combined Hamiltonian $H_{\rm combined} = V_{eN} + \omega_{\rm magn}$ is hermitian by linearity, and the Pauli sum is additive. The $\omega_{\rm magn}$ contribution is suppressed relative to V_{eN} by the dimensionless ratio $r_Z/a_0 \sim 2 \times 10^{-5}$. This realizes Phase B of the multi-focal sprint sequence's structural expectation that $\omega_{\rm magn}$ is W1a's inner-fluctuation sibling: it shares the joint-register layout, the diagonal-density JW encoding, and the multi-λ Shibuya–Wulfman infrastructure with V_{eN}, differing only in the radial kernel ($-Z/|\mathbf{r}_e - \mathbf{R}_p|$ vs. ρ_M).

The structural reading is that $\omega^{\rm comp} = \omega_{\rm recoil} + \omega_{\rm magn} + \cdots$ is the multi-focal-composition analog of the Connes–Chamseddine inner fluctuation $\omega = \omega_{\rm gauge} + \omega_{\rm Higgs}$ on the SM almost-commutative spectral triple [39, 45]: each component is built from a fixed parameterization of the

proton's internal structure (ρ_E for $\omega_{\rm recoil}$, ρ_M for $\omega_{\rm magn}$), the framework computes the operator action on the joint Hilbert space, and the choice of profile family and calibration scalars is Layer-2 input from QCD/scattering data, in the sense of Paper 18 §IV's inner-factor input data tier [40].

D. Test on deuterium 1S hyperfine structure

The composed nuclear–electronic architecture of this section is designed for $I = 1/2$ proton hyperfine structure (the 21 cm hydrogen test case of Sec. VI). A natural extension is to $I = 1$ deuteron HFS: the cross-register V_{eN} machinery is unchanged (Roothaan 1951 multipole expansion is angular, independent of nuclear spin), the magnetization-density operator $\omega_{\rm magn}$ is parameterized by a scalar moment $r_Z(D)$, and only the hyperfine-coupling Hamiltonian $A_{\rm hf}\,\hat{\mathbf{I}}\cdot\hat{\mathbf{S}}$ acquires a different Clebsch–Gordan multiplicity factor $(2I+1)/2 = 3/2$ between the $F = I + 1/2$ and $F = I - 1/2$ levels.

The framework-native Bohr–Fermi prediction at strict point-nucleus, no-recoil order is

$$\nu_{\rm HFS}^{\rm BF,strict}(D,1S) = \frac{3}{2}\times\frac{4}{3}\,\frac{g_d^{\rm atomic}}{2}\,\alpha^2\left(\frac{m_e}{m_p}\right)\mathrm{Ha}, \quad (18)$$

where $g_d^{\rm atomic} = 2\mu_d/\mu_N = 1.7148$ is the atomic-physics Hamiltonian-convention g-factor (twice the CODATA value $g_d = \mu_d/(\mu_N\cdot I) = 0.857$ because of the factor $2I$); the divisor restoring $\mu_I/(I\mu_N)$ is 2, not $2I$ — the two coincide only at $I = 1$. The mass factor is m_e/m_p for every nucleus, because the nuclear magneton $\mu_N = e\hbar/(2m_p)$ is defined with the proton mass. With CODATA constants this yields 327.2350 MHz vs the Wineland–Ramsey 1972 [48] measurement of 327.384352522(2) MHz, a residual of −456 ppm. (Corrected 2026-08-28: this equation previously carried m_e/m_d in place of m_e/m_p with no factor-of-2 divisor. Since $m_d/m_p = 1.99900750 \approx 2$ the two errors nearly cancelled, leaving the baseline high by 496.5 ppm; $I = 1/2$ nuclei were unaffected, so the H and T cross-checks did not expose it.)

The leading-order Zemach correction with $r_Z(D) = 2.593$ fm (Friar–Sick [49]) gives $\Delta\nu_Z/\nu_F = -98.0$ ppm at the operator level via `geovac.magnetization_density.hydrogen_zemach_eides_leading_order`, matching the analytic Eides $-2Zm_e r_Z$ to floating-point precision. Profile (Gaussian vs. exponential) independence is preserved at leading order, identical to the hydrogen regression of Sec. VII C. The cumulative chain (BF + recoil + Schwinger a_e + leading Zemach) lands at 327.3153 MHz with −211 ppm residual against experiment. The chain omits the deuteron's nuclear-structure corrections beyond the leading Zemach term (the inelastic, polarizability contribution among them) plus multi-loop QED (LS-8a wall, ∼few ppm) plus recoil NLO and finite-size charge corrections; the residual is not apportioned among them here.

a. Multi-focal architecture verdict at $I = 1$. The Bohr–Fermi residual of −456 ppm is the cleanest framework-native verification of the $I = 1$ cross-register angular coupling structure in the catalogue. The Hamiltonian $A_{\rm hf}\hat{\mathbf{I}}\cdot\hat{\mathbf{S}}$ is unchanged between $I = 1/2$ and $I = 1$; only the Clebsch–Gordan multiplicity differs. The deuteron's nonzero electric quadrupole moment $Q_d = 0.286$ efm^2 couples to electronic charge-density gradients but enters HFS at sub-ppm level for s-states (no orbital angular momentum to couple to) and is negligible at the precision tested here. At leading order the multi-focal architecture handles $I = 1$ identically to $I = 1/2$, and the only sub-leading mechanism that genuinely opens up at $I = 1$ (the deuteron quadrupole) sits below the Layer-2 input threshold for s-state observables.

Sprint provenance: `debug/precision_catalogue_deuterium_hfs` `debug/precision_catalogue_deuterium_hfs_memo.md`, `debug/data/precision_catalogue_deuterium_hfs.json`. Operator-level four-component Roothaan autopsy (May 9 2026): `debug/d_hfs_autopsy_track5_memo.md`, which elevates the $I = 1$ verdict from energy level to operator level via explicit construction of $\hat{I}\cdot\hat{S}$ on the 6-dim joint nuclear-electronic Hilbert space (multiplicity ratio $D/H = 3/2$ at machine precision) and reproduction of the Eides analytic leading-order Zemach scalar at $\sim 1.45\times10^{-14}$% of the LO shift via the production magnetization-density operator at $r_Z(D) = 2.593$ fm. Cross-references: Paper 34 §V (catalogue row for $\nu_{\rm HFS}(D)$ at −456 ppm BF strict and −211 ppm full chain), Paper 34 §V.C.X (`sec:autopsy_d_hfs` operator-level autopsy), Sprint MH Track B (electronic→muonic mass swap at unchanged proton; this sprint is the spin-1/2→spin-1 nucleus swap at unchanged electron), Sprint HF Track 4 (hydrogen r_Z Layer-2 input, 0.012% Eides regression).

E. Synthesis and forward references

The cross-register V_{eN} and $\omega_{\rm magn}$ operators together close the operator-level scaffolding of two of the multi-focal walls (W1a and W1b) named in the May 2026 sprint sequence. The honest verdicts are scope-bounded:

- V_{eN}: leading-order Bethe–Salpeter recoil to 2.86%, its scale fixed by calibrating λ_n; sub-leading content basis-truncation-bounded at $n_{\max} = 1$, with the 2.86% drift identified as the half-integer-mass-ratio artifact tower (structurally absent in Pachucki 2018). Multi-shell $n_{\max} \geq 2$ on both registers is the structural follow-up for higher-order recoil.

- $\omega_{\rm magn}$: the closed-form leading-order Zemach term (Eides § 11.1.1), with ρ_M calibrated to $M_1 = r_Z$, profile independence at leading order and shape dependence appearing only at the second moment $\langle r^2\rangle_{\rho_M}$. Higher-order Eides corrections (polarizability $\Delta_{\rm pol}$, recoil-corrected Zemach, Friar

moment $\langle r^3 \rangle_{(2)}$) each enter as additional inner-fluctuation components in the same architectural slot.

a. Cross-references. The convergence theorem for the single-factor Camporesi–Higuchi truncation of the S^3 spectral triple (van Suijlekom state-space Gromov–Hausdorff, commutative factor) is Paper 38 [41]; it is context for, not a derivation of, the electron register used here. The tensor-product extension — the two-infinite-factor convergence theorem adjacent to the cross-register operator construction reported here — is Paper 39 [42]. The standalone NCG-positioning observation for the Track NI base construction, including the structural literature gap against the published Connes–Marcolli–Chamseddine and Connes–van Suijlekom lineages, is the Track NI Zenodo memo [47].

b. Open follow-ups. Three downstream sprints are named but not pursued in this paper: (i) multi-shell $n_{\max} \geq 2$ on both joint registers, to close the next-order sign disagreement against Pachucki 2018 [46] and lift the basis-truncation ceiling on V_{eN}; (ii) full Pachucki $(Z\alpha)^4$ match in the multi-shell extension, requiring the integer-power coefficients to agree term-by-term rather than only at leading order; (iii) n_e-projected FCI on the joint register including the Slater–Condon decomposition needed for many-electron antisymmetrization, which is the structural extension of the diagonal-density encoding used here. Each is architecturally compatible with the present construction; each is a scoped follow-up sprint.

VIII. DISCUSSION

The results of Sections II–VI suggest a natural partition of the GeoVac framework into a universal component and a Coulomb-specific component:

- **Universal (applies to HO, Woods–Saxon, NN potentials).** The angular selection rules from Wigner $3j$ symbols and the Gaunt integral; the angular-sparsity reduction of single-center Pauli scaling below the naive $O(Q^4)$ (measured $O(Q^{3.79})$ for Coulomb atoms — a narrow margin under the exact global-M_L rule) and the exactly linear composed Pauli scaling in Q at fixed basis; the block structure of the two-body tensor; the qubit encoding via Jordan–Wigner on species-specific registers.

- **Coulomb-specific.** The S^3 conformal projection $p_0 = \sqrt{-2E_n}$; the SO(4) hidden symmetry and the Runge–Lenz vector; the $-1/16$ kinetic constant; the S^3 spectral data entering the fine-structure constant observation [32].

The nuclear shell model fits squarely in the universal category. The Fock rigidity theorem makes this precise: the conformal projection simply does not survive the transition from Coulomb to HO or Woods–Saxon, so any claim that the GeoVac S^3 machinery extends directly to nuclei would be incorrect. What does extend is the angular machinery — and the results of Sections IV and V demonstrate that this machinery is sufficient to produce structurally sparse qubit Hamiltonians for simple nuclei.

Directions for follow-up work. (i) Larger N_{shells} for converged binding energies; the interesting physics question is how the Pauli count grows with Q as the basis grows, and whether the Minnesota potential's short range eventually fills in the angular zeros. (ii) Three-body forces (chiral EFT at N^{2}LO); these introduce new classes of Pauli terms beyond the two-body pattern studied here, and their effect on sparsity is an open question. (iii) Larger nuclei such as ^{6}Li, ^{12}C, and ^{16}O; the composed architecture should apply, with nucleon blocks labelled by single-particle (n_r, ℓ, j) orbitals. (iv) Hardware implementation and comparison with NCSM benchmarks on actual quantum simulators; the 16-qubit deuteron is within reach of current ion-trap and superconducting platforms.

A final note on possible Coulomb-free projections. The Bargmann–Segal transform [17] gives a natural holomorphic representation of the HO Hilbert space, in which the HO Hamiltonian becomes a first-order differential operator. Its discrete label lattice on S^5 is built in Paper 24 [38]: the HO spectrum sits on the labels as a diagonal, and the lattice's node Laplacian L_0 is spectrally inert — the same situation as the S^3 label lattice, whose edge Laplacian does not converge to Δ_{S^3} (Paper 7).

IX. CONCLUSION

We have constructed qubit Hamiltonians for the nuclear shell model on the GeoVac hyperspherical lattice. The HO shell closures and Mayer–Jensen magic numbers are reproduced from the single-particle spectrum with standard spin–orbit and orbital-correction parameters. The deuteron and He-4 qubit Hamiltonians, built from the Minnesota potential via Moshinsky–Talmi brackets, reproduce their full-CI spectra; at the same 16 qubits He-4 has 31% more Pauli terms than the deuteron, its added pp and nn channels, while its 12.25× larger Hilbert space costs no terms. The composed nuclear–electronic encoding, with proton-spin hyperfine and proton-radius couplings, demonstrates that the multi-block architecture extends to nucleon–electron systems but faces a $\sim 10^{13}$ coefficient ratio that requires block-partitioned simulation. The Fock projection rigidity theorem precisely delineates which parts of the GeoVac framework transfer to non-Coulomb systems (the angular machinery) and which do not (the S^3 conformal projection); a finite nucleus keeps the electron's labels, releases n from integrality and consumes moments of its charge as matching coefficients (for the hyperfine factor, the Zemach moment first), and past $Z\alpha = 1$ makes the radius compulsory (Sec. III A). The binding energies reported here at $N_{\text{shells}} = 2$ are not converged and should not be

read as nuclear-structure predictions; they are encoding-validation benchmarks that verify the qubit Hamiltonian reproduces the matrix representation.

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